

Manan Singh

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EDUCATION

Purdue University

B.S. in Robotics Engineering Technology, Minor in Computer Science — GPA: 3.55

West Lafayette, IN

Aug. 2023 – May 2027

EXPERIENCE

Production Engineering Intern

August 2025 - December 2025

Toyota Motor Manufacturing Indiana

Princeton, IN

- Applying Toyota Production System and 5S philosophies to upgrade and improve Toyota Car paint booths

Systems Test Engineer Intern

May 2025 – August 2025

Cubic Transportation Systems

San Diego, CA

- Executed 50+ test cases to validate and report defects found within the Chicago Transit fare collection system
- Integrated a robot arm with a desktop automation framework (PyWinAuto) and wrote 5 automated tests, increasing smoke testing efficiency while adding the ability to test transit card taps
- Wrote 15 automated tests in Java to verify and validate transit system OpenAPI functionality
- Wrote a web server and motor control program on a microcontroller to remotely control a cycle tester for bus validators, adding variable speed control and enabling more realistic test cases to be executed

Lead Mechanical Engineer

Aug. 2023 – May 2025

Railside Robotics

West Lafayette, IN

- Guided and mentored a team of 60 club members through an iterative design cycle, teaching top-down CAD modeling, rapid prototyping, and DFM techniques, culminating in the construction of 20 combat robots that have competed in over 60 matches across three semesters
- Led the design, manufacturing, and construction of a modular and serviceable 128 ft³ steel arena weighing 1500lbs to replace an outdated and unsafe 32 ft³ arena while conducting cost analysis, leading to a budget surplus of \$3,000
- Implemented club-wide Product Data Management (PDM) system (Aras Innovator) and onboarded 60 members in storing CAD files and accessing standard part files through PDM to streamline collaboration and design review
- Represented Purdue University and Railside Robotics as a student delegate at the Aras ACE 2024 Conference and presented the club's PLM process to 100+ corporate attendees, leading to the reception of a \$10,000 grant

Printing & Prototyping Lab Peer Mentor

Oct. 2024 – Present

Bechtel Innovation Design Center

West Lafayette, IN

- Teaching students to operate metal and nonmetal laser cutters as well as FDM, SLA, and SLS 3D printers
- Following SOPs to provide manufacturing services and complete machine maintenance/upkeep
- Developed a lab shift-scheduling program that integrated with the Google Calendar API and center reservation system to create optimized work schedules for myself and my co-workers
- Mentored 30+ students in design, prototyping, and manufacturing processes

PROJECTS

Culverizer | Combat Robot

April 2022 – Present

- Designed and manufactured a robust 3-lb battlebot leveraging manufacturing techniques involving laser cutting, 3D printing, waterjet cutting, and milling that won 15 of 23 battlebot matches
- Optimized weapon blade geometry, strength, and weight with FEA to build a 1,000+ Joule weapon system capable of incapacitating other robots instantaneously while maintaining durability and repairability

AWARDS AND CERTIFICATES

- 2022 FIRST Robotics Competition World Champion (team254.com/first/2022)
- OSHA 10 Certification
- Rising Boilermaker Award (Purdue University Office of Student Life, 2025)
- Finite Element Analysis Milestone Certificate (Purdue University, 2023)

SKILLS

Mechanical: Solidworks, Siemens NX, Fusion 360, FEA, PDM, DFM, CAM

Software: C/C++, Java, Python, PLC Programming